

GUJARAT POLICE INNOVATION CHALLENGE 2026 · SOLUTION PRESENTATION

Sentinel

An open, explainable interoperability & intelligence layer for
Gujarat's 80,000+ camera, 26-department CCTV network

Model 1 + Model 2 Hybrid · Vehicle-Tracking Test Case · Geo-Temporal Identity Resolution

Prize Pool ₹51,00,000 · Phase 1 Deadline 7 Sep 2026 · Category A

PROBLEM UNDERSTANDING

26 departments. 80,000+ cameras. Zero shared visibility.

Gujarat's CCTV infrastructure is fragmented across analog/IP cameras, multiple vendors and VMS platforms, cloud and local storage, and 7–15+ day retention windows — spread across border districts, Valsad, Dahod, Somnath, Jamnagar, Dwarka and beyond, ~1,000 km apart.

The core ask

- Integrate heterogeneous departmental CCTV into one ecosystem, without replacing existing systems
- Correlate live feeds with government databases (VAHAN, SARTHI, eGujCop, AFIS, NAFIS) for automated alerts
- Scale toward ~80,000 cameras, statewide

The mandatory test case

- Onboard ~50 geographically distributed, heterogeneous cameras
- Given a **vehicle registration number**, trace it across the network
- Produce a **timestamped, location-wise movement history**
- Cross-reference a watchlist and fire **real-time alerts** on match

PROPOSED SOLUTION MODEL & JUSTIFICATION

Hybrid Model 1 + Model 2, with a documented Model 3/4 evolution path

Model 1 — built REAL

Central CCTV registry + GIS mapping. The common inventory every other model sits on top of.

Model 2 — built REAL

Unified viewing, direct RTSP/ONVIF connection — no new middleware, existing departmental VMS keeps running untouched.

Model 3/4 — roadmap DOCUMENTED

VMS federation + statewide central AI platform. Cited as the production evolution, **not claimed as built**.

Why this, not a from-scratch platform

Gujarat already operates VISWAS/NETRAM/TRINETRA at scale (7,000+ cameras, Phase-II adding ~10,500 more). Pitching a from-scratch centralized CCTV platform invites the reasonable question: "*what did you invent that doesn't already exist?*" Model 2's direct-connection approach is both what the problem statement recommends for teams not building a federation bus, and the most honestly-scoped target achievable in the build window.

This is a hybrid in the rules' own definition (§7: "combine elements from two or more models"), and an explicit bonus criterion. We claim hybrid credit only for what is genuinely built and verified — not for the Model 3/4 elements, which stay on the roadmap.

Not another ANPR system — an explainable identity-resolution layer

Detection, ANPR, and cross-camera tracking are already commercially shipped in India — Staqu (JARVIS), Videonetics, Vehant, and Innefu, among others. This submission does not claim to have invented those.

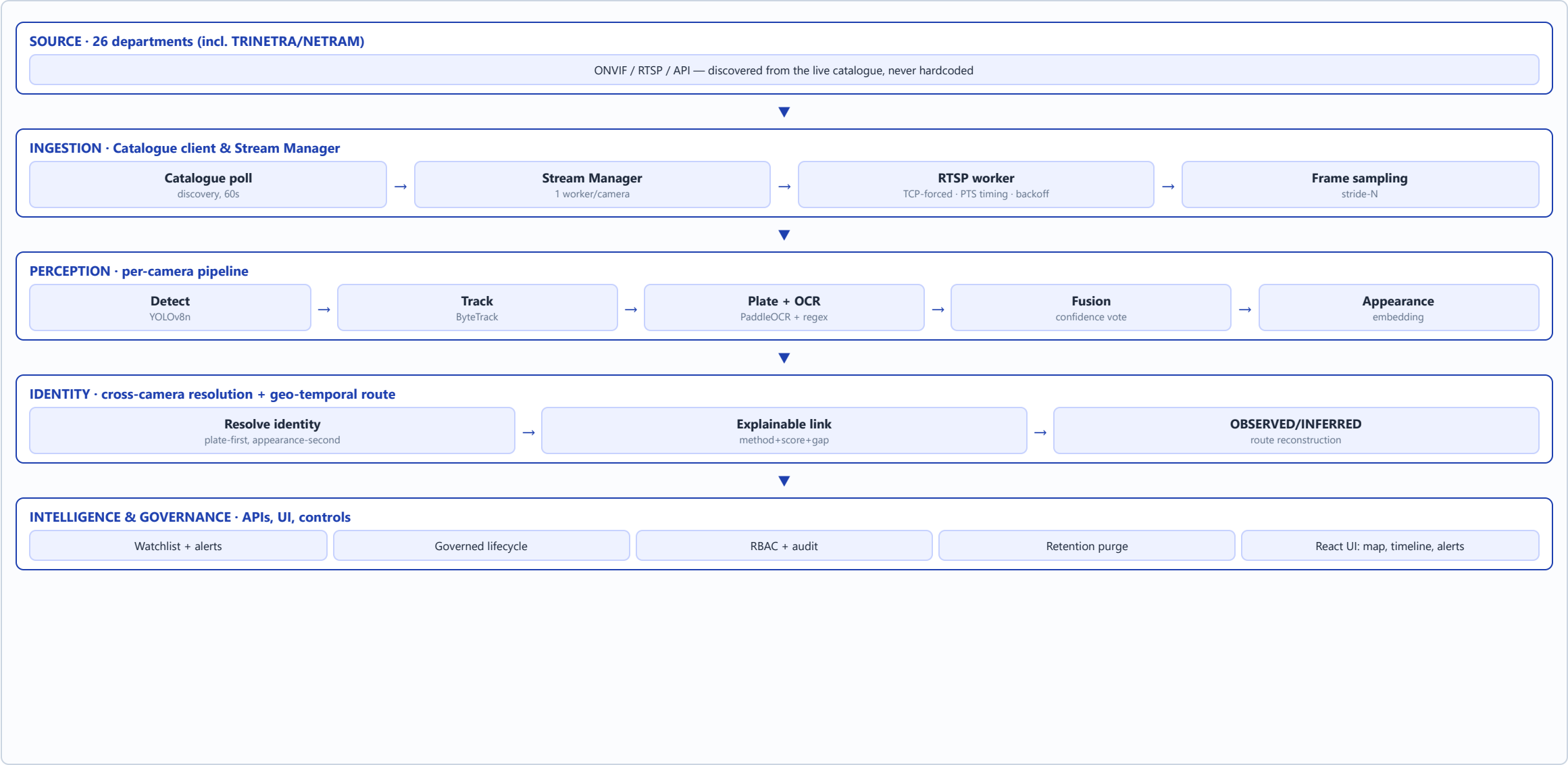
Sentinel's genuine contribution: a **persistent vehicle entity** assembled from uncertain observations — plate-OCR confidence, appearance similarity, temporal & geographic feasibility — where every cross-camera link records **why** it was made, and an unreadable plate degrades gracefully to attribute/appearance tracking instead of losing the vehicle.

Four things that differentiate us

- **ANPR failure ≠ tracking failure** — appearance + attributes carry identity through plate-read gaps, upgrading retroactively the moment any camera reads the plate
- **Geo-temporal route reconstruction** — confirmed sightings + honestly-labelled *inferred* movement across camera blind-spots (not "tracked")
- **Explainable linking** — every match shows plate/appearance/recency evidence, never a black-box score
- **Governed alert lifecycle** — new → acknowledged → resolved/dismissed, with an accountable false-positive path, not a fire-and-forget alert

HIGH-LEVEL SYSTEM ARCHITECTURE

Five layers: source → ingestion → perception → identity → intelligence



From a registration number to a courtroom-ready timeline

The investigator's path

1. Enter a plate + purpose + case ID (purpose-bound, audited)
2. System returns every sighting across the network — including cameras where the plate was never read
3. Route rendered as confirmed sightings + inferred segments between them
4. Each link shows its evidence: plate confidence, appearance similarity, time consistency
5. Any watchlist match already fired an alert — investigator works it through acknowledge → resolve/dismiss

Behind the scenes, per camera

1. Sampled frame → YOLOv8 vehicle detection → ByteTrack assigns a persistent track
2. Plate region localized, enhanced (CLAHE), read via PaddleOCR, validated against the Indian plate regex
3. Reads voted across the whole track (temporal fusion) — a single bad frame can't corrupt the result
4. Low-confidence reads withheld — vehicle still logged by colour/type/appearance
5. Identity resolver matches this sighting to an existing vehicle, or starts a new one

Detection, recognition & event analytics — status stated honestly

Stage	Approach	Status
Vehicle detection	YOLOv8n, pretrained COCO (car/truck/bus/motorcycle/person)	REAL / VERIFIED
Within-camera tracking	ByteTrack (Ultralytics built-in), one persistent tracker per camera	REAL / VERIFIED
Plate OCR	PaddleOCR + regex Indian-plate validation + temporal vote fusion	REAL / VERIFIED (SYNTHETIC TEXT)
Plate localization	Heuristic crop + CLAHE enhancement	WORKING SUBSTITUTE
Appearance Re-ID	HSV colour histogram + shape template, cosine similarity	REAL / WORKING
Vehicle attributes	Colour, type, dimensions — the fallback identifier when ANPR fails	REAL / WORKING
Motion analytics	Dwell time, speed, direction from track path over PTS — enables wrong-way/stopped-zone filters	REAL / VERIFIED
Cross-camera identity	Plate-first, appearance-second resolution with retroactive plate upgrade	REAL / VERIFIED END-TO-END
Geo-temporal route	OBSERVED sightings + honestly-labelled INFERRED camera-less segments	REAL / VERIFIED
Make/model classification	No credible open-source Indian-market model exists	HONEST GAP

Honest accuracy note: OCR verified against clean synthetic plate text and real-photo detection — real Indian-plate footage accuracy is not yet measured. Field accuracy comes from detector + regex validation + temporal fusion working together, not the OCR engine alone.

Geo-temporal route reconstruction: honest about what was actually seen



Why this matters for a police tool

- A dashed segment is never presented as "tracked" — it is an **inference**, labelled as such, with its basis shown
- Straight-line (haversine) distance + time-gap → a plausibility check; distance is a lower bound, so a high implied speed is flagged as "**not a direct drive**" — an unmonitored detour, or possibly a different vehicle
- This directly targets the official **bonus criterion**: "advanced cross-camera vehicle movement tracking / multi-camera correlation"

Scope discipline: full road-network (OSM) routing and predictive next-camera pruning are **roadmap**, not built — OSM coverage of Gujarat's minor roads is incomplete, and pruning buys nothing observable at 50-camera demo scale. Its value is the 80,000-camera scalability story.

Continuous cross-reference, with a governed lifecycle — not a fire-and-forget toast

Correlation

Every persisted vehicle sighting with a resolved plate is checked against the watchlist (stolen/wanted/blacklisted) at write time. A match creates an alert immediately — no batch delay.

Production integration targets

VAHAN, SARTHI, eGujCop, CCTNS — design principle is **query, don't copy**: Sentinel queries these systems for a match on demand rather than replicating their data.

Governed alert lifecycle (Palantir-inspired entity lifecycle)



Only declared transitions are permitted (illegal moves rejected with HTTP 409); every step records who acted and when — the concrete expression of "AI output is a lead, not evidence."

"Why was this vehicle linked?" — never a black box

Every cross-camera link records

- **Link method** — new identity / plate continuation / plate upgrade / appearance match
- **Re-ID similarity score** — cosine similarity, when appearance carried the link
- **Time gap** — seconds since the identity's previous sighting

Fused transparency score

Plate confidence 50% + appearance similarity 35% + recency 15% — explicitly labelled a **transparency aid, not a calibrated probability**, so an investigator's judgement is never replaced by a number.

Why this is a jury-relevant safeguard

- Counters **automation bias** — low-confidence, appearance-only links are visually distinguished, not equal-weighted with a plate-confirmed one
- Known failure mode stated, not hidden: two similar-coloured/model vehicles can produce a false appearance-only match — which is exactly why plate evidence always outranks appearance
- No route in the system auto-dispatches an officer or auto-confirms an identity — every action is a human decision

Real controls, not a compliance checklist

Access control & audit

- Ranked RBAC — viewer < investigator < admin — every route enforces its minimum role (401/403, not logged-and-allowed)
- API keys stored SHA-256 hashed, never plaintext
- Every query is **purpose-bound and audit-logged** against the authenticated user (not a self-reported string — closes a real spoofing gap found and fixed during the build)
- Found and fixed: an SSRF vulnerability in the HLS proxy — same-host validation added, verified against both the attack and the legitimate case

DPDP & data governance

- Retention **enforced in code** (30-day pilot default for events; 365-day for audit logs — modelled on, and improving on, the UK ANPR system's criticised lack-of-enforcement)
- Data minimization — attribute-only fallback instead of storing raw video
- Purpose limitation — no generic "browse all activity"; every query needs a plate, case, or watchlist match to start from
- Face recognition deliberately **out of scope** — DPDP risk + documented demographic bias, and not required by the mandatory vehicle test case

Centralize events, not video — the only credible path to 80,000 cameras

80,000 cameras × ~2 Mbps ≈ **160 Gbps sustained, ~1.7 PB/day** if raw video were centralized. Not credible for any team at this budget — so Sentinel centralizes structured events instead: ≈1 KB/event × 80,000 cameras ≈ **640 Mbps** at full scale — a three-orders-of-magnitude reduction.

Tier	Role	Sizing
Edge	RTSP ingestion, YOLOv8+PaddleOCR inference, within-camera tracking	~20 cameras/GPU (mobile-scale models) → ~4,000 GPU-class units at full build-out
Regional	Cross-camera Re-ID within region, regional media relay	Aligned to Gujarat's 34 existing NETRAM centres, not duplicating them
Central	Cross-region identity, watchlist, RBAC, GIS registry, dashboards	Same FastAPI+SQLAlchemy code as this submission, pointed at PostgreSQL

Phased rollout

Pilot (this submission, ~30 real cameras) → district pilot (PostgreSQL, one edge unit) → regional (Kafka event bus) → statewide (~80,000 cameras). **Same codebase at every phase** — nothing here is throwaway.

HA/DR & low-bandwidth

PostgreSQL primary/replica; regional stores replicate async so a regional outage doesn't lose central watchlist/RBAC. Low-connectivity districts (Valsad, Dahod, Dwarka) run detection/OCR at the edge unconditionally — only KB-scale events cross the link.

Open source throughout, per the hackathon's Open Source Requirement

Backend & AI

- **FastAPI** — RBAC-gated REST API
- **SQLAlchemy** — SQLite (dev) → PostgreSQL/PostGIS-ready, idempotent additive migrations
- **OpenCV** — RTSP ingest, enhancement, colour/embedding math, geo-temporal math
- **Ultralytics (YOLOv8 + ByteTrack)** on PyTorch
- **PaddleOCR 2.9.1** on PaddlePaddle (version-pinned — verified 3.0+ breaks the wrapper)

Frontend & delivery

- **React + Vite** — Map/Registry, Vehicle Tracking, Watchlist, Registry Ops tabs
- **Leaflet** — Model 1 GIS map, health-colour-coded
- **hls.js** — live view via authenticated backend HLS proxy
- Evaluated public models named honestly: IISc UVH-26 (Indian-CCTV detector), Awiros ANPR-OCR — cited as verified, licensed upgrade paths, **not integrated** this build (version-compatibility / provenance risk)

What this submission does not yet claim

Gap	Why it's stated, not hidden
Real Indian-plate OCR accuracy unmeasured	Only verified on clean synthetic text + real-photo detection; the single most jury-visible number, honestly flagged as unmeasured against live footage
Plate localization is heuristic, not a fine-tuned model	No fine-tuned Indian-plate localizer was available in the build window; documented upgrade path (real, licensed candidates evaluated) exists
Department-scoped RBAC not built	Flat role set today; schema (<code>ApiKeyEntry.department</code>) already supports the extension without restructuring
WHEP low-latency preview not built	HLS via authenticated proxy is the working live-view path
Live sandbox validation unconfirmed in latest session	Reconfirm before relying on it for the live demo, rather than assume from an earlier session
Model 3/4 (federation, central VMS) not built	Deliberate — a half-built federation bus is worse than an honest documented roadmap

Why we lead with this: this is a government/police evaluation. A visible, stated gap is a smaller risk than an overstated capability discovered live.

EXPECTED OPERATIONAL BENEFITS & IMPACT

From manual footage-hunting to a searchable, explainable intelligence layer

For investigators

A registration number produces a complete cross-department timeline in seconds — replacing manual, camera-by-camera footage requests across independently-run departmental systems.

For command centres

Real-time watchlist alerts with a governed lifecycle mean nothing is silently missed or silently over-trusted — every hit is triaged and accounted for.

For the state

An interoperability layer that **extends** existing investment (TRINETRA/NETRAM, departmental VMS) rather than requiring a rip-and-replace — the realistic path to statewide coverage.

Policing impact

- Faster vehicle-of-interest tracing across jurisdictional/departmental boundaries
- Reduced false-accusation risk via explainable, confidence-scored links and mandatory human confirmation
- A governance-first design reduces legal/DPDP exposure relative to blanket-surveillance approaches

Public safety impact

- Faster stolen-vehicle recovery and missing-vehicle response
- A foundation (Model 1 registry) that also serves infrastructure planning and gap-analysis, independent of the AI layer

Sentinel

An open, explainable, honestly-scoped interoperability layer —
built on what's real, roadmapped on what isn't.

[Technical Proposal \(HLD\)](#) · [Plan for Scale](#) · [Live & own-feed demonstrations](#) · [Source available on request](#)